الأستاذ: عقيلة راف الله عقيلة

(Lecture 2)

Gamma Camera

A **Gamma Camera** is a medical imaging device used in **nuclear medicine** to visualize the distribution of radioactive tracers (within the patient's body).

It is also known as:

- **Anger Camera**
- **Scintillation Camera**

The device was invented by **Hal Anger** in 1958.

Radiopharmaceuticals

Most Important Radiopharmaceuticals Used in Gamma Camera

1. Technetium-99m (Tc-99m) (Most important and most widely used)

Used in:

- Bone scan
- Cardiac scan
- Renal scan
- Liver scan
- Thyroid scan

Why is Tc-99m the best?

Emits **gamma radiation only**

Has → **optimal energy (140 keV)**

Short half-life (**6 hours**)

Safe for the patient

2. Iodine-123 (I-123)

📌 **Used in:**

- Thyroid imaging
- Thyroid function studies

Emits **gamma radiation only**

Has a suitable half-life

3. Iodine-131 (I-131)

Used in:

- Thyroid imaging
- Treatment of thyroid cancer

Note:

It emits:

- Gamma radiation
- Beta radiation

4. Thallium-201 (Tl-201)

Used in:

- Cardiac imaging
- Myocardial perfusion studies

5. Gallium-67 (Ga-67)

Used in:

- Tumors
- Infections

6. Indium-111 (In-111)

Used in:

- Infection imaging
- White blood cell studies

Most Commonly Used Radiopharmaceutical

Technetium-99m (Tc-99m)

Used in more than:

80% of Gamma Camera procedures

Main Components of a Gamma Camera

A Gamma Camera consists of the following essential components:

- Collimator
- Scintillation Crystal
- Photomultiplier Tubes (PMTs)
- Electronics
- Computer System

1. Collimator

Definition

A **collimator** is a device made of **lead** or **tungsten** and is positioned in front of the detector.

Function

- Allows gamma rays traveling in specific directions to pass through
- Absorbs oblique (angled) photons
- Improves image resolution

Only **parallel photons** reach the detector.

Types of Collimators

1. **Parallel-Hole Collimator**
    - The most commonly used type
2. **Pinhole Collimator**
    - Used for imaging small organs such as the **thyroid gland**

Factors Affecting Collimator Performance

- Hole length (L)
- Hole diameter (d)
- Septal thickness

Resolution Equation

$$R = \frac{d(l + H)}{l}$$

Where:

- d = hole diameter
- L = hole length
- H = distance from the patient

2. Scintillation Crystal

Definition

A **scintillation crystal** is a material that converts **gamma radiation into visible light photons**.

Common Material

The most widely used material is:

NaI (Tl) (Thallium-activated Sodium Iodide)

Where:

- NaI = Sodium Iodide
- Tl = Thallium

Crystal Thickness

Typically ranges from **6 to 12 mm**.

Working Principle

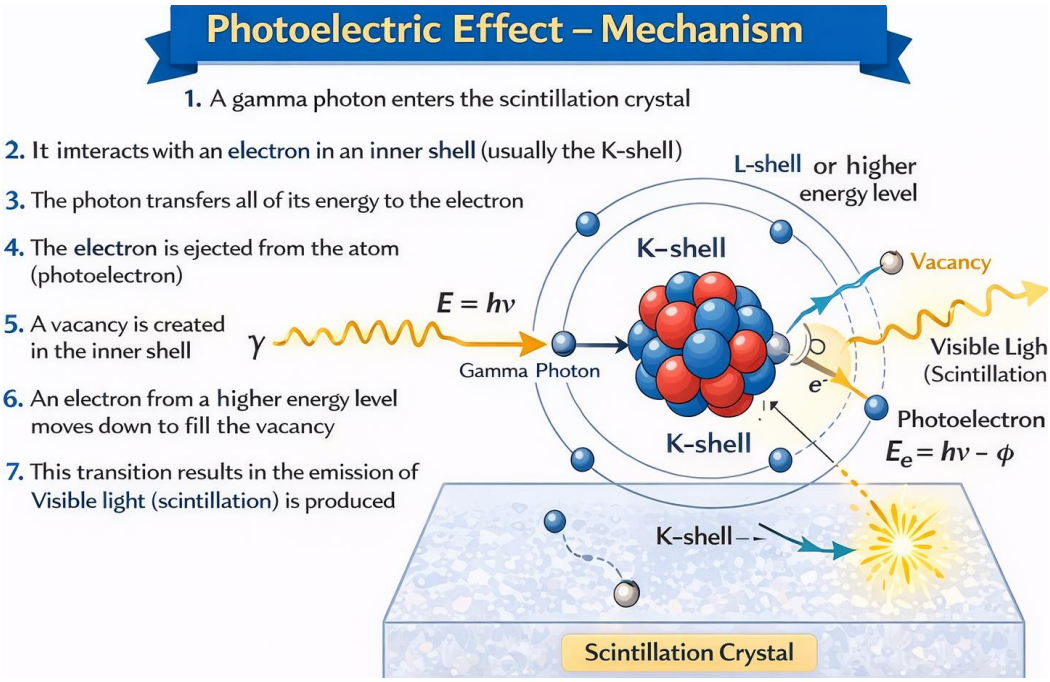
1. Gamma rays enter the crystal
2. They interact with the crystal material by Photoelectric effect

Photoelectric Effect – Mechanism

The **photoelectric effect** occurs through the following steps:

1. A gamma photon enters the scintillation crystal

2. It interacts with an electron in an inner shell (usually the K-shell)
3. The photon transfers all of its energy to the electron
4. The electron is ejected from the atom (photoelectron)
5. A vacancy is created in the inner shell
6. An electron from a higher energy level moves down to fill the vacancy
7. This transition results in the emission of Visible light (scintillation) is produced



Why NaI(Tl) is Used

1. High Efficiency

Effectively absorbs gamma rays

- Has a relatively high atomic number
    - Na = 11

- I = 53

2. High Light Output

- A single gamma photon produces **thousands of light photons**
- This facilitates detection by the **photomultiplier tubes (PMTs)**

What Happens After the Crystal?

- The emitted light is detected by **Photomultiplier Tubes (PMTs)**
- The light signals are converted into **electrical signals**
- These signals are processed by the electronics and computer system to form an image

Photomultiplier Tubes (PMTs)

Number

Approximately **30–100 tubes** are used.

Function

- Detect incoming light photons
- Convert light into an electrical signal
- Amplify the electrical signal

Conversion of Light into Electricity

The first part in the PMT is called the:

Photocathode

Its function:

- Converts photons (light) into electrons

This process is called:

Photoelectric effect

Signal Amplification

This is the most important part

The PMT contains:

a series of electrodes called **Dynodes**

How does amplification occur?

1. One electron is emitted from the **Photocathode**
2. It reaches the first **Dynode**
3. Several electrons are produced (e.g., 3–5)
4. These move to the second **Dynode**
5. They multiply again

This process repeats multiple times

Result:

The signal is amplified thousands or even millions of times

Final Stage

The electrons finally reach the:

Anode

Where they are converted into:

a measurable electrical current

Why do we need a PMT?

Because:

The light produced by the crystal is very weak

We need to amplify it so it can be detected and analyzed



